

NERF Erasure-Rate Monotonicity

Literature-Already-Answered Identification

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Status. `literature_already_answered`. This note records an exact later answer; it is not claimed as a new discovery.

1 Original question

Let $F = [f_1 \cdots f_N]$ be an $M \times N$ frame. Fickus and Mixon call F a (p, C) -numerically erasure-robust frame (NERF) if every submatrix $F_{\mathcal{K}}$ with $|\mathcal{K}| = K = (1 - p)N$ satisfies $\text{Cond}(F_{\mathcal{K}}) \leq C$.

In Section 4, source-PDF page 14, they ask whether a (p, C) -NERF must also be a (p', C) -NERF for every $p' \in [0, p)$ for which $L = (1 - p')N$ is integral [1]. Equivalently, does the same worst-case condition-number bound persist when more columns survive?

2 Exact later answer

Liu explicitly cites the Fickus–Mixon question in the introduction of [2]. Lemma 1 (pp. 586–587) proves

$$\max_{|\mathcal{K}|=K} \text{Cond}(F_{\mathcal{K}}) \geq \max_{|\mathcal{S}|=L} \text{Cond}(F_{\mathcal{S}}) \quad (K < L).$$

Therefore every (p, C) -NERF is a (p', C) -NERF whenever $p' < p$ and the two prescribed cardinalities are integral. This is exactly the affirmative answer requested in the source.

3 Independent verification

For completeness, fix a set $\mathcal{S}$ of L columns and write $A_{\mathcal{T}} = F_{\mathcal{T}} F_{\mathcal{T}}^*$ for each K -element subset $\mathcal{T} \subset \mathcal{S}$. If $\text{Cond}(F_{\mathcal{T}}) \leq C$, then

$$\lambda_{\max}(A_{\mathcal{T}}) \leq C^2 \lambda_{\min}(A_{\mathcal{T}}).$$

The cone defined by this inequality is closed under finite sums: for any such positive semidefinite matrices A_j ,

$$\begin{aligned} \lambda_{\max} \left(\sum_j A_j \right) &\leq \sum_j \lambda_{\max}(A_j) \leq C^2 \sum_j \lambda_{\min}(A_j) \\ &\leq C^2 \lambda_{\min} \left(\sum_j A_j \right). \end{aligned}$$

Every column of $F_{\mathcal{S}}$ occurs in exactly $\binom{L-1}{K-1}$ of the K -subsets, and hence

$$\sum_{\substack{\mathcal{T} \subset \mathcal{S} \\ |\mathcal{T}|=K}} A_{\mathcal{T}} = \binom{L-1}{K-1} F_{\mathcal{S}} F_{\mathcal{S}}^*.$$

The preceding cone inequality and invariance under positive scalar multiples give $\text{Cond}(F_{\mathcal{S}})^2 \leq C^2$. Since $\mathcal{S}$ was arbitrary, F is a (p', C) -NERF. This also independently checks the decisive content of Liu's Lemma 1.

4 Scope and search record

The identification resolves the monotonicity question only. It does not address the source's separate questions asking whether NERFs exist for $p \geq \frac{1}{2}$ or what the largest admissible erasure rate is.

A bounded novelty search used the exact source title, the phrases “deleting columns does not necessarily worsen” and “ (p', C) -NERF”, and NERF/erasure-rate monotonicity terminology. It found Liu's 2020 paper, whose introduction explicitly says it answers the Fickus–Mixon question and whose Lemma 1 gives the exact comparison.

References

- [1] M. Fickus and D. G. Mixon, *Numerically erasure-robust frames*, Linear Algebra and its Applications 437 (2012), 1394–1407; arXiv:1202.4525.
- [2] Y. Liu, *Comparison on the robustness against erasure rates of numerically erasure-robust frames*, International Journal of Applied Mathematics 33 (2020), 585–590, doi:10.12732/ijam.v33i4.3.